RingBuffer

What is it ?

- A ring buffer is a data structure where operations like pushing, popping, shifting, and unshifting are O(1), using modulo arithmetic to wrap around an array. It maintains order by using head and tail indices that can move circularly within a fixed-size array.

What happen when it need to resize ?

- When a ring buffer needs to resize, it creates a new larger buffer, starting at the head and copying elements in order. The head will be set to 0, and the tail will be set to the current length, allowing for additional capacity and continued circular operations.

Use case ?

- A ring buffer can be used in log batching scenarios, where logs need to maintain order while being written. It allows for efficient logging by periodically flushing a batch of logs without blocking the main service, and without requiring complex mutex synchronization.

Object pool

- An object pool is a technique for reusing objects instead of creating new ones repeatedly, which can improve performance and memory usage.
- While ring buffers can be used for object pooling, a simple ArrayList is often sufficient if the order of object creation is not important.